COMSATS University Islamabad, Lahore Campus

Midterm Examination – Semester Fall 2021

Course Title:	Database Systems-I			Course Code:	CSC371	Credit	4 (3, 1)
Course Instructor/s:	Dr. Hamid Turab Mirza			Programme Name:	BSCS		
Semester:		Batch	SP20-BCS	Section	B and C	Date:	15 November 2021
Time	**90 Minutes**			**Maximum Marks:**		**50**	

Part A

Q 1: Explain the term Metadata? Also discuss its significance in database systems. **[3+7 = 10 Marks]**

Q 2: What is the purpose of cardinality constraints? List four types of cardinality constraints and elaborate anyone with an example. **[2+4+4 = 10 Marks]**

Q 3: What is an identifier attribute? State three criteria for selecting identifiers for entities. **[2+3 = 05 Marks]**

Q 4: Explain each of the following, elaborate with example **[2.5+2.5 = 05 Marks]**

A: Derived attribute

B: Composite attribute

Compiled by Hafiz Ahmad
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Part B

Q 5: ERD **[20 Marks]**

A General Hospital consists of a number of specialized wards (such as Maternity, Surgery, Oncology, etc). Each ward hosts a number of patients, who were admitted on the recommendation of their own family doctor and confirmed by a consultant employed by the Hospital. On admission, the personal details of every patient are recorded. A separate register is to be held to store the information of the tests undertaken and the results of a prescribed treatment. A number of tests may be conducted for each patient. Each patient is assigned to one leading consultant but may be examined by another doctor, if required. Doctors are specialists in some branch of medicine and may be leading consultants for a number of patients, not necessarily from the same ward.

Draw an ER diagram to represent the data requirements as following:

- Identify the main entity types. **[8]**
- Identify the main relationship types between the entity types. **[8]**
- Identify attributes and associate them with entity or relationship types. **[2]**
- Determine candidate and primary key attributes for each entity type. **[2]**